

1. (a) The least upper bound: 3
The greatest lower bound: -1
- (b) The least upper bound: Doesn't exist.
The greatest lower bound: 1
- (c) The least upper bound: Doesn't exist.
The greatest lower bound: Doesn't exist.
- (d) The least upper bound: 0
The greatest lower bound: -1
2. (a) $\frac{n+3}{\ln(n+3)}$ is increasing since $\frac{x}{\ln x}$ is an increasing function on $[4, \infty)$.
And it is bounded below by $\frac{4}{\ln 4}$ but not bounded above.
- (b) $\frac{(-1)^n}{n}$ is not monotonic.
And it is bounded below by -1 and bounded above by $\frac{1}{2}$.
- (c) Set $a_n = \frac{3^n}{(n+1)^2}$. Then $\frac{a_{n+1}}{a_n} = 3 \left(\frac{n+1}{n+2} \right)^2 > 1$.
So $\frac{3^n}{(n+1)^2}$ is increasing.
And it is bounded below by $\frac{3}{4}$ but not bounded above.
- (d) $\frac{n^2}{\sqrt{n^3+1}} = \frac{1}{\sqrt{\frac{1}{n} + \frac{1}{n^4}}}$ and $\frac{1}{n} + \frac{1}{n^4}$ decreases to 0 $\Rightarrow \frac{n^2}{\sqrt{n^3+1}}$ is increasing.
And it is bounded below by $\frac{\sqrt{2}}{2}$, but not bounded above.
3. Boundedness: $0 < (c^n + d^n)^{1/n} < (2d^n)^{1/n} = 2^{1/n}d \leq 2d$
Monotonicity:

$$\begin{aligned}
 a_{n+1}^{n+1} &= c^{n+1} + d^{n+1} \\
 &= cc^n + dd^n \\
 &< (c^n + d^n)^{1/n} c^n + (c^n + d^n)^{1/n} d^n \\
 &= (c^n + d^n)^{1+1/n} \\
 &= (c^n + d^n)^{(n+1)/n} \\
 &= a_n^{n+1}
 \end{aligned}$$

Taking the $(n+1)$ st root of each side we have $a_{n+1} < a_n \Rightarrow$ The sequence is decreasing.

4. (a) $\left(1 - \frac{1}{n}\right)^n$ converges to e^{-1} as $n \rightarrow \infty$.
Since $\left(1 - \frac{1}{n}\right)^n = \left(1 + \frac{(-1)}{n}\right)^n \rightarrow e^{-1}$ as $n \rightarrow \infty$.
- (b) $\int_0^n e^{-nx} dx$ converges to 0 as $n \rightarrow \infty$.
Since $\int_0^n e^{-nx} dx = -\frac{e^{-n^2}}{n} + \frac{1}{n} \rightarrow 0$ as $n \rightarrow \infty$

(c) If $0 < t < 1$, $(|t| + \frac{x}{n})^n$ converges to 0 as $n \rightarrow \infty$.

If $t = 1$, $(|t| + \frac{x}{n})^n$ converges to e^x as $n \rightarrow \infty$.

If $t > 1$, $(|t| + \frac{x}{n})^n$ diverges as $n \rightarrow \infty$.

Since $(t + \frac{x}{n})^n = t^n \left(1 + \frac{x/t}{n}\right)^n$.

(d) $\frac{\arctan n}{n}$ converges to 0 as $n \rightarrow \infty$.

$$5. \sqrt{n+1} - \sqrt{n} = \frac{\sqrt{n+1} - \sqrt{n}}{\sqrt{n+1} + \sqrt{n}} (\sqrt{n+1} + \sqrt{n}) = \frac{1}{\sqrt{n+1} + \sqrt{n}}$$

$$\lim_{n \rightarrow \infty} \sqrt{n+1} - \sqrt{n} = \lim_{n \rightarrow \infty} \frac{1}{\sqrt{n+1} + \sqrt{n}} = 0.$$

$$6. \sqrt{n^2 + n} - n = \frac{\sqrt{n^2 + n} - n}{\sqrt{n^2 + n} + n} (\sqrt{n^2 + n} + n) = \frac{n}{\sqrt{n^2 + n} + n} = \frac{1}{1 + \sqrt{1 + 1/n}}$$

$$\lim_{n \rightarrow \infty} \sqrt{n^2 + n} - n = \lim_{n \rightarrow \infty} \frac{1}{1 + \sqrt{1 + 1/n}} = \frac{1}{2}$$

7. Let $\epsilon > 0$.

If $a_n \rightarrow L$, then there exists a positive integer k such that $|a_p - L| < \frac{\epsilon}{2}$ for all $p \geq k$

With $m, n \geq k$, we have

$$|a_m - a_n| \leq |a_m - L| + |L - a_n| = |a_m - L| + |a_n - L| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$$

Hence, every convergent sequence is a Cauchy sequence.

8. (a) Since $\{a_n\}$ converges, it is a Cauchy sequence,

so given $\epsilon > 0$, we can find k such that $|a_n - a_m| < \epsilon$ for $n, m \geq k$.

In particular, $|a_{n+1} - a_n| < \epsilon$, so $\lim_{n \rightarrow \infty} (a_n - a_{n-1}) = 0$

(b) $\{a_n\}$ does not necessarily converge.

For example, let $a_n = \ln n$ and it diverges,

but $a_n - a_{n-1} = \ln n - \ln(n-1) = \ln \left(\frac{n}{n-1} \right) \rightarrow \ln 1 = 0$ as $n \rightarrow \infty$.

9. Let M be a bound for $\{b_n\}$. Then $|a_n b_n| \leq |a_n| M$.

Given $\epsilon > 0$, choose k such that $|a_n| < \frac{\epsilon}{M}$ for $n \geq k$.

Then $|a_n b_n| < \epsilon$ for $n \geq k$. Therefore, $a_n b_n \rightarrow 0$ as $n \rightarrow \infty$.

10. Proof:

Since the absolute-value function is everywhere defined and everywhere continuous, then $a_n \rightarrow L$ implies $|a_n| \rightarrow |L|$

The converse is false.

For example, let $a_n = (-1)^n$.

Then $|a_n| \rightarrow 1$ as $n \rightarrow \infty$, but $\{a_n\}$ diverges.

11. By the continuity of f , $f(L) = f(\lim_{n \rightarrow \infty} a_n) = \lim_{n \rightarrow \infty} f(a_n) = \lim_{n \rightarrow \infty} a_{n+1} = L$.

12. (a) $\lim_{x \rightarrow \infty} \frac{\ln(x^k)}{x} = \lim_{x \rightarrow \infty} \frac{k/x}{1} = 0$
- (b) $\lim_{x \rightarrow \infty} \left(1 + \frac{a}{x}\right)^{bx} = \lim_{x \rightarrow \infty} \left[\left(1 + \frac{a}{x}\right)^x\right]^b = (e^a)^b = e^{ab}$
- (c) Since $\lim_{x \rightarrow \infty} \ln \left[(x^3 + 1)^{1/\ln x}\right] = \lim_{x \rightarrow \infty} \frac{\ln(x^3 + 1)}{\ln x} = \lim_{x \rightarrow \infty} \frac{\left(\frac{3x^2}{x^3+1}\right)}{1/x} = \lim_{x \rightarrow \infty} \frac{3}{1 + 1/x^3} = 3$
 thus, $\lim_{x \rightarrow \infty} (x^3 + 1)^{1/\ln x} = e^3$
- (d) $\lim_{x \rightarrow 0} (x \ln |\sin x|) = \lim_{x \rightarrow 0} \frac{\ln |\sin x|}{1/x} = \lim_{x \rightarrow 0} \frac{\frac{\cos x}{\sin x}}{-\frac{1}{x^2}} = \lim_{x \rightarrow 0} \left(-\frac{x}{\sin x}\right) (x \cos x) = 0$
- (e) $\lim_{x \rightarrow \infty} x \sin \frac{\pi}{x} = \lim_{t \rightarrow 0^+} \frac{\sin \pi t}{t} = \lim_{t \rightarrow 0^+} \frac{\pi \cos \pi t}{1} = \pi$
- (f) $\lim_{x \rightarrow 1} \left(\frac{1}{\ln x} - \frac{x}{x-1}\right) = \lim_{x \rightarrow 1} \frac{x-1-x \ln x}{(x-1) \ln x} = \lim_{x \rightarrow 1} \frac{-\ln x}{(x-1)(1/x) + \ln x}$
 $= \lim_{x \rightarrow 1} \frac{-x \ln x}{x-1+x \ln x} = \lim_{x \rightarrow 1} \frac{-\ln x - 1}{2 + \ln x} = -\frac{1}{2}$
- (g) $\lim_{n \rightarrow \infty} \frac{1}{n} \ln \frac{1}{n} = \lim_{n \rightarrow \infty} -\frac{\ln n}{n} = 0$
- (h) $\lim_{n \rightarrow \infty} \frac{\ln n}{n^p} = \lim_{n \rightarrow \infty} \frac{1/n}{pn^{p-1}} = \lim_{n \rightarrow \infty} \frac{1}{pn^p} = 0$
- (i) $\lim_{x \rightarrow 1} \frac{\ln x}{1-x} = \lim_{x \rightarrow 1} \frac{1/x}{-1} = -1$
- (j) $\lim_{x \rightarrow 0} \frac{\sqrt{a+x} - \sqrt{a-x}}{x} = \lim_{x \rightarrow 0} \frac{\frac{1}{2\sqrt{a+x}} + \frac{1}{2\sqrt{a-x}}}{1} = \frac{1}{\sqrt{a}} = \frac{\sqrt{a}}{a}$
- (k) $\lim_{x \rightarrow 0} \frac{\tan^{-1} x}{\tan^{-1} 2x} = \lim_{x \rightarrow 0} \frac{1}{1+x^2} \frac{1+4x^2}{2} = \frac{1}{2}$
- (l) $\lim_{x \rightarrow 0} \frac{\sin^{-1} x}{x} = \lim_{x \rightarrow 0} \frac{\frac{1}{\sqrt{1-x^2}}}{1} = 1$
- (m) $\lim_{x \rightarrow 0} \frac{a^x - (a+1)^x}{x} = \lim_{x \rightarrow 0} \frac{a^x \ln a - (a+1)^x \ln(a+1)}{1} = \ln \left(\frac{a}{a+1}\right)$